

DYNAMIC COMPATIBILITY RECOMMENDATION SYSTEM

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this report does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma in any other university or institution of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where acknowledgement is made in the text.

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ABSTRACT

The agricultural sector heavily depends on machinery for efficient land preparation, cultivation, harvesting, irrigation, and transportation. However, agricultural machinery maintenance is often affected by the difficulty of identifying compatible spare parts, especially when original equipment manufacturer parts are unavailable, expensive, or delayed. This individual research contribution focuses on the development of a Dynamic Compatibility Recommender as part of the Smart Spare Parts Management System for Agricultural Machinery. The proposed component is designed to recommend compatible and functionally equivalent spare parts by combining structured compatibility rules, semantic similarity, machine learning prediction, and feedback-aware ranking. A structured compatibility schema is introduced using attributes such as machine family, compatibility group, function type, substitute level, confidence score, and evidence source. These attributes allow spare parts to be evaluated not only by textual similarity but also by mechanical and functional compatibility. The recommendation process is implemented through hard constraints, soft constraints, vector-based similarity ranking, Random Forest compatibility prediction, and a weighted hybrid scoring mechanism. The Random Forest classifier uses pairwise part features such as category match, machine-family match, compatibility-group match, function-type match, material similarity, diameter difference, price difference, lifespan difference, and existing similarity score to produce a compatibility probability. In addition, an explanation builder is included so that each recommendation can present matched fields, key differences, substitute level, and confidence score. The component is evaluated through recommender ranking metrics such as Precision@5 and Recall@5, and classification metrics such as accuracy, precision, recall, and F1-score. The expected outcome is a scalable, explainable, and practically useful recommender that reduces incorrect purchases, improves spare-part discovery, supports farmer decision-making, and strengthens vendor inventory planning in agricultural machinery spare-part ecosystems.

Keywords: Dynamic compatibility recommender, agricultural machinery, spare parts, hybrid recommendation, Random Forest, semantic similarity.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CSV	Comma-Separated Values
DB	Database
F1	F1-score
HP	Horsepower
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
ML	Machine Learning
NLP	Natural Language Processing
OEM	Original Equipment Manufacturer
RF	Random Forest
UI	User Interface
UX	User Experience

1. INTRODUCTION

The Smart Spare Parts Management System for Agricultural Machinery is designed to address practical problems faced by farmers, machinery owners, repair workshops, and spare-part vendors when searching for suitable replacement parts. Agricultural machinery is used in demanding environments, and delays in replacing damaged or worn components can directly affect cultivation schedules, harvesting activities, machine availability, and farmer income. In many agricultural contexts, the spare-part selection process is still heavily dependent on manual inspection, vendor experience, and incomplete product descriptions. This creates a significant risk of incorrect purchases, repeated workshop visits, and unnecessary downtime. The Dynamic Compatibility Recommender is developed as the individual research component that focuses specifically on solving this compatibility and recommendation problem.

The key purpose of the component is to recommend spare parts that are not only textually similar, but also mechanically and functionally compatible. A basic search engine may return parts that share similar names or descriptions; however, agricultural machinery spare parts require stronger compatibility logic. For example, a belt, filter, bearing, tyre, blade, or gear component may look similar in name but differ in dimensions, material, mounting type, function, operating load, or compatible machine family. Therefore, the recommendation process must consider structured specifications, machine relationships, functional roles, and substitute quality. This requirement motivates the development of a dynamic compatibility recommender rather than a simple keyword-based search module.

1.1 Background and Literature Survey

Recommendation systems are software techniques that provide suggestions for items likely to be useful to a user. The literature commonly categorizes recommendation methods into collaborative filtering, content-based filtering, knowledge-based recommendation, and hybrid recommendation approaches [1], [2]. Collaborative filtering depends on user-item interaction patterns, while content-based filtering relies on

item features and similarity between item descriptions [1], [14]. In spare-part recommendation, however, user interaction histories are often sparse, and the most important decision factor is not popularity but compatibility. Therefore, the Dynamic Compatibility Recommender is positioned closer to a knowledge-based and hybrid recommendation system, where domain rules, item attributes, and machine-learning predictions are combined [3], [4].

Hybrid recommender systems have been widely discussed as a way to overcome the limitations of using a single recommendation method. Burke explains that hybrid systems can combine multiple recommendation strategies to improve robustness and reduce weaknesses such as cold-start limitations and over-specialization [3]. Adomavicius and Tuzhilin also emphasize that future recommender systems require richer contextual and domain knowledge to improve recommendation quality [1]. This is directly relevant to agricultural spare parts because compatibility cannot be decided only by previous user behavior. A user may never have purchased a particular substitute part before, but the system must still be able to recommend it when the specifications and machine relationships are valid.

Knowledge-based recommendation is particularly important for domains where explicit constraints and expert knowledge must be considered [4], [33]. In agricultural machinery spare parts, knowledge may include machine family, compatibility group, function type, part category, dimensional limits, material suitability, lifespan expectations, and evidence source. These relationships allow the system to make recommendations based on engineering logic instead of only statistical similarity. For this reason, the proposed component introduces a structured compatibility schema to represent the domain knowledge required for safe spare-part recommendation.

Content-based and semantic similarity methods remain useful because spare-part records include text fields such as name, brand, category, machine model, description, and specification details. Vector-space models and cosine similarity have traditionally been used in information retrieval to compare item representations [5], [6]. In the proposed

component, the similarity pipeline is used to rank candidates after rule-based filtering, allowing the system to use both descriptive information and structured attributes. This prevents the system from relying only on exact keyword matches and supports flexible search behavior for farmers and vendors who may use informal part names.

Machine learning has also been applied in many recommendation and classification tasks because it can learn patterns from data and produce predictions for unseen cases [7], [8], [9]. Random Forest is selected for this component because it is suitable for tabular features, can model non-linear relationships, and is relatively interpretable compared with more complex black-box methods [7]. In this research, the Random Forest model is trained using pairwise spare-part data, where each row represents a pair of parts and a compatibility label. Pairwise features such as same category, same machine family, same compatibility group, same function type, same material, diameter difference, price difference, lifespan difference, and similarity score are used to predict compatibility probability.

Explainability is another important research consideration. Explainable recommendation aims to provide not only accurate suggestions but also understandable reasons for why an item is recommended [10]. In the agricultural spare-part domain, a farmer or vendor must be able to understand whether a recommended part is exact, near, or functional substitute. Without explanations, users may not trust the system or may misunderstand compatibility risks. Therefore, the proposed component includes an explanation builder that outputs matched fields, differences, substitute level, confidence score, and a short recommendation reason.

Evaluation of recommender systems is commonly conducted using top-N ranking metrics such as Precision@K and Recall@K, which measure how many relevant recommendations appear within the top positions [11], [12]. For the machine learning classifier, standard classification metrics such as accuracy, precision, recall, and F1-score are used [8], [13]. This dual evaluation approach is suitable because the component has two tasks: ranking compatible spare parts and classifying pairwise

compatibility probability. In addition, baseline comparisons such as rules-only, similarity-only, similarity with feedback, Random Forest-assisted, and final hybrid versions are used to demonstrate improvement over simpler approaches.

In relation to agricultural machinery, literature on mechanization emphasizes that access to suitable machinery and maintenance services affects the productivity and sustainability of farming systems [22], [23]. A practical spare-parts management solution must therefore support timely maintenance, reduce downtime, and improve part availability. The proposed Dynamic Compatibility Recommender contributes to this broader need by improving spare-part discovery and substitute identification for agricultural machinery users.

Reference / Approach	Main Technique	Strength	Limitation Identified	Relevance to Proposed Component
Collaborative filtering recommenders	User-item interaction patterns	Effective when sufficient user history exists	Cold-start and sparse data issues in spare-part domain	Shows why interaction-only recommendation is unsuitable
Content-based recommenders	Item attribute similarity	Useful for textual and specification similarity	May ignore mechanical compatibility constraints	Used as part of semantic/vector similarity ranking
Knowledge-based recommenders	Explicit domain rules and constraints	Suitable for technical products with constraints	Requires structured domain knowledge	Foundation for machine family, compatibility group, and function type
Hybrid recommenders	Combination of multiple methods	Improves robustness and accuracy	Requires careful score normalization	Adopted through similarity, rules, ML, and feedback
Explainable recommenders	Recommendation with reasons	Improves transparency and trust	Needs well-designed explanation logic	Implemented through matched fields, differences, and confidence
Proposed Dynamic Compatibility Recommender	Rule-based filtering + vector ranking + Random Forest + feedback	Supports mechanical compatibility and explainability	Requires reliable part data and final testing	Addresses the identified agricultural spare-part compatibility gap

Table 1.1: Comparison of existing approaches and identified research gaps

1.2 Research Gap

The main research gap identified from the literature and domain analysis is the lack of an intelligent, compatibility-aware, and explainable spare-part recommendation mechanism for agricultural machinery. Existing general-purpose recommendation methods can suggest similar items, but they do not sufficiently represent the mechanical and functional relationships that determine whether one spare part can safely replace another. Agricultural spare parts require compatibility decisions based on category, function, machine family, compatibility group, dimensions, material, durability, price, and user acceptance. Therefore, a simple recommender that only compares product names or user clicks is not adequate.

The second gap is the limited use of structured substitute levels and confidence explanations. A farmer may need to know whether a substitute part is an exact replacement, a near substitute with minor differences, or a functional substitute that can work under certain conditions. Most simple recommendation systems do not communicate this distinction clearly. The proposed component addresses this gap through substitute level, confidence score, evidence source, and explanation generation.

The third gap is the lack of hybrid integration between rule-based safety, semantic similarity, machine learning probability, and user feedback. Rule-based methods are safe but may be too rigid. Similarity methods are flexible but may recommend invalid items. Machine learning can learn complex patterns but requires data and validation. Feedback improves practical ranking but should not override safety constraints. The proposed system combines these approaches into a structured hybrid scoring mechanism.

1.3 Research Problem

The research problem addressed in this individual component can be stated as follows: Farmers and spare-part vendors lack an intelligent and explainable mechanism for dynamically identifying compatible spare parts for agricultural machinery when original parts are unavailable, expensive, or difficult to identify. This problem causes incorrect purchasing decisions, unnecessary repair delays, increased maintenance cost, and

reduced operational reliability of machinery. Therefore, a dynamic compatibility recommender is required to evaluate spare parts using structured domain rules, semantic similarity, machine learning probability, and feedback-aware ranking.

1.4 Research Objectives

1.4.1 Main Objective

The main objective of this research is to design and develop a Dynamic Compatibility Recommender that recommends compatible and functionally equivalent spare parts for agricultural machinery using a hybrid approach based on compatibility rules, semantic similarity, Random Forest prediction, and user feedback.

1.4.2 Specific Objectives

1. To design a structured compatibility schema using machine family, compatibility group, function type, substitute level, confidence score, and evidence source.
2. To validate and standardize spare-part data fields such as name, brand, category, machine model, description, specifications, diameter, price, lifespan, and material.
3. To implement hard and soft compatibility rules that filter unsafe candidates and refine ranking quality.
4. To integrate vector-based similarity ranking for semantic comparison of part records.
5. To construct a pairwise dataset and train a Random Forest classifier for compatibility prediction.
6. To implement a hybrid recommendation score combining similarity score, rule score, Random Forest probability, and feedback score.
7. To generate explanations for recommendations using matched fields, key differences, substitute level, and confidence score.
8. To evaluate recommender performance using Precision@5 and Recall@5 and evaluate the classifier using accuracy, precision, recall, and F1-score.

2. METHODOLOGY

2.1 Understanding the Key Pillars of the Research Domain

2.1.1 Recommendation Systems

A recommendation system is a decision-support mechanism that suggests relevant items to users based on available data, user requirements, item properties, or domain rules. In the proposed component, the item being recommended is an agricultural machinery spare part. Unlike media or e-commerce recommendation, the recommended item must satisfy mechanical compatibility. Therefore, the system uses a hybrid approach rather than relying solely on collaborative filtering. The recommender must understand the part category, function, machine model, and technical attributes before ranking alternatives.

2.1.2 Knowledge-Based and Rule-Based Reasoning

Knowledge-based reasoning is required because spare-part compatibility is governed by explicit relationships. For example, a tyre replacement must match or be acceptable for a machine family and wheel dimension, while a belt must match function type, section type, and length range. Rule-based reasoning is used to separate hard constraints from soft constraints. Hard constraints are conditions that must be satisfied before a candidate can be recommended. Soft constraints influence the score without fully excluding the candidate. This approach improves safety while preserving flexibility.

2.1.3 Semantic Similarity and Vector Representation

Semantic similarity allows part records to be compared using text and attribute information. Each part can be represented using a feature vector generated from fields such as name, brand, category, machine model, description, and specifications. Similarity ranking can then identify candidates that are close to the selected part. The vector pipeline is useful because farmers and vendors may use different names for similar parts, and an exact keyword match may fail to capture real similarity. However, semantic similarity is applied after compatibility rules to prevent invalid matches.

2.1.4 Machine Learning Prediction

Machine learning is used to predict compatibility probability between a selected part and a candidate part. Random Forest is selected because it performs well on structured tabular data and can combine categorical match indicators with numeric differences. The model learns from pairwise examples labelled as compatible or incompatible. Its probability output is then used as one part of the final hybrid score, rather than as the only decision factor.

2.1.5 Explainable Recommendation

Explainability is essential in this research because users must understand why a recommended part is suitable. A farmer may not trust a substitute recommendation unless the system explains that the machine family, compatibility group, and function type match, while the diameter difference is within an acceptable threshold. Therefore, each recommendation includes `why_recommended`, `matched_fields`, `differences`, `substitute_level`, and `confidence`. This improves transparency and supports practical decision-making.

2.2 Overall Approach

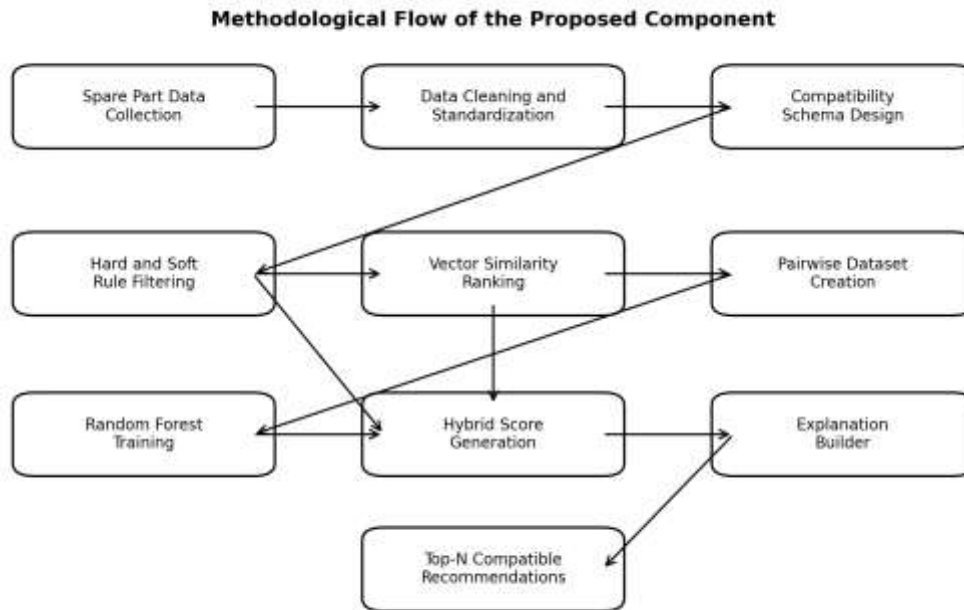


Figure 2.1: Methodological flow of the Dynamic Compatibility Recommender

As shown in Figure 2.1, the methodology begins with spare-part data collection and data cleaning. The next step is the design of a compatibility schema that represents machine family, compatibility group, function type, substitute level, confidence score, and evidence source. After the schema is introduced, hard and soft rules are applied to control candidate selection. Vector similarity is then used to rank candidates according to descriptive and specification-level similarity. A pairwise dataset is created to train the Random Forest classifier, and the classifier probability is incorporated into the hybrid score. The final output is a ranked list of compatible spare parts with explanations.

2.3 Data Collection and Data Preparation

The data required for the Dynamic Compatibility Recommender consists of agricultural machinery spare-part records, machine model information, technical specifications, and substitute relationship indicators. The base spare-part record includes the part name, brand, category, machine model, description, specifications in JSON format, diameter, material, price, and expected lifespan. These fields are required because compatibility depends on both textual identity and measurable technical characteristics. The data preparation process includes checking missing values, removing duplicate part records, validating numeric fields, standardizing machine model names, and correcting inconsistent material or category names.

Data cleaning is treated as a major methodological step because recommendation quality depends heavily on input quality. If a machine model is misspelled or if a diameter value is missing, the recommender may incorrectly exclude a valid candidate or include an invalid candidate. Therefore, the system standardizes values before scoring. Important fields are checked to ensure that name, category, machine_model, description, specs_json, diameter, price, lifespan, and material are usable. This step also supports the vector rebuild process by ensuring that meaningful text and numeric attributes are available for similarity calculation.

For machine learning, the original part records are converted into pairwise records. A pairwise record contains Part A, Part B, and a compatibility label. Positive pairs are created from known compatible or substitute relationships, while negative pairs are created from parts with different functions, incompatible categories, or unrelated machine families. This dataset is required because the Random Forest model predicts compatibility between pairs rather than classifying a single part.

2.4 Compatibility Schema Design

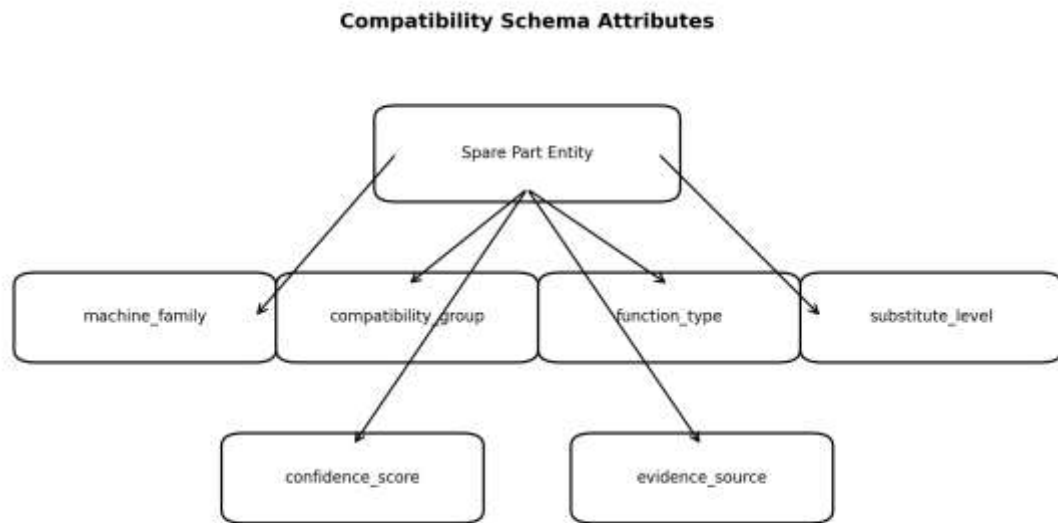


Figure 2.2: Compatibility schema attributes

The compatibility schema is the main structural improvement introduced in this research. The previous base system could compare similarity, but it did not fully represent compatibility relationships. The improved schema introduces fields that allow the system to behave as a compatibility-aware recommender rather than a simple similarity engine. These fields make substitute relationships explicit and allow the recommendation API to return clear evidence to the user.

Field	Description	Example	Purpose in Recommender
-------	-------------	---------	------------------------

machine_family	A broader group of machines that share similar operational or structural characteristics.	Medium Tractor Series 45-60 HP	Used as a hard compatibility rule and RF feature.
compatibility_group	A technical group that links interchangeable or comparable parts.	VBELT_A_SECTION_54	Used to identify valid substitutes.
function_type	The mechanical function served by the part.	Power Transmission	Prevents functionally incorrect recommendations.
substitute_level	The quality of substitution between original and candidate.	exact / near / functional	Communicates replacement risk and suitability.
confidence_score	A normalized value representing confidence in recommendation.	0.86	Displayed to user and used in ranking/explanation.
evidence_source	The basis used to justify compatibility.	Manufacturer catalog / expert input / field feedback	Improves traceability and trust.

Table 2.1: Compatibility schema fields

The use of machine family ensures that parts are not recommended across unrelated machines. Compatibility group provides the core linking key for interchangeable parts. Function type ensures that parts serve the same mechanical purpose. Substitute level communicates whether the recommendation is exact, near, or functional. Confidence score supports ranking and user interpretation. Evidence source adds traceability by indicating whether the relationship is based on manufacturer specification, expert recommendation, field usage, or system-calculated similarity.

2.5 Rule-Based Filtering

The rule-based component is designed to make the recommender safer and easier to justify. The rules are divided into hard rules and soft rules. Hard rules must be satisfied before a part is allowed to enter the candidate set. Soft rules are used to influence ranking after the hard rules are passed. This separation is important because some compatibility requirements are strict, while others are comparative.

2.5.1 Hard Rules

- The candidate must belong to the same or compatible category as the selected part.
- The candidate must have the same function type.
- The candidate must belong to a compatible machine family.
- The candidate must have a valid compatibility group.
- The selected original part itself must be excluded from the final recommendation list.

Hard rules are applied before similarity and machine learning ranking. This ensures that the recommendation engine does not rank candidates that are mechanically invalid. For example, a part with high textual similarity but a different function type is removed before the scoring stage.

2.5.2 Soft Rules

- Diameter closeness is used when the part has dimension-based compatibility.
- Material closeness is used to compare durability and operating suitability.
- Price closeness or price advantage is used to support cost-aware recommendation.
- Lifespan closeness is used to avoid recommending parts with significantly weaker durability.
- Feedback acceptance is used to promote candidates that have been accepted by users in previous interactions.

Soft rules do not remove all candidates by themselves. Instead, they adjust the score. For example, a substitute part with the same function type and compatibility group may remain valid even if the price differs, but the price difference is shown in the explanation and may influence ranking.

2.6 Similarity-Based Ranking

The similarity ranking component compares the selected part with candidate parts using a vector-based similarity pipeline. The vector is generated from selected fields such as part name, brand, category, machine model, description, and technical specifications. After the hard rules are applied, the similarity score ranks candidates based on closeness

of representation. This is useful in the spare-part domain because parts may have similar roles but different names or brands. The vector representation allows the system to identify these relationships more flexibly than keyword matching.

The similarity score is not used alone because semantic closeness does not guarantee mechanical compatibility. For example, two belts may have similar descriptions but different sections or lengths. Therefore, similarity ranking is placed after hard filtering and before hybrid score generation. This order supports both safety and flexibility.

2.7 Random Forest Compatibility Prediction

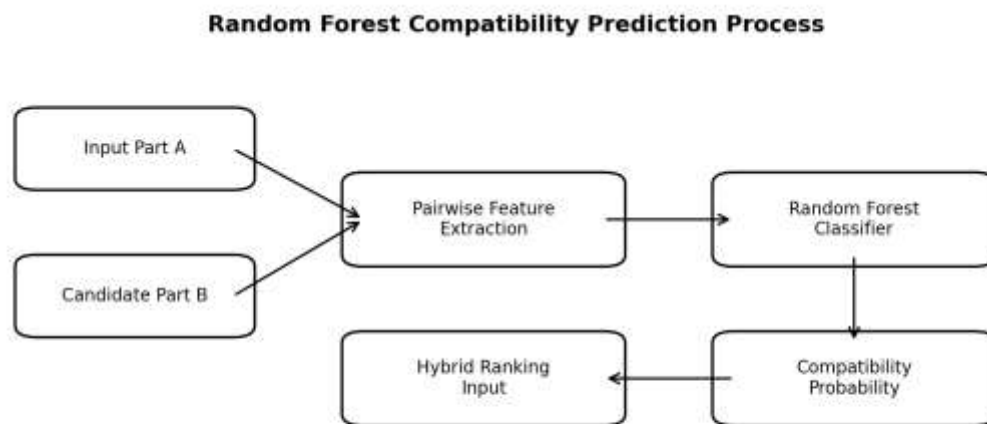


Figure 2.3: Random Forest compatibility prediction process

The machine learning component uses a Random Forest classifier to predict compatibility probability for each candidate pair. The model is trained using a pairwise dataset in which each row represents two spare parts and a compatibility label. The classifier does not replace rule-based reasoning; instead, it supports the hybrid score by providing a learned probability based on historical or labelled compatibility patterns.

2.7.1 Pairwise Dataset Construction

The pairwise dataset is created by generating positive and negative pairs. Positive pairs represent known compatible or substitute parts. Negative pairs represent incompatible pairs, such as different categories, different machine families, unrelated function types, or invalid dimensions. This structure allows the model to learn the relationship between two parts.

2.7.2 Feature Engineering

The Random Forest model uses both binary match features and numeric difference features. Binary features indicate whether two parts share the same category, machine family, compatibility group, function type, and material. Numeric features capture diameter difference, price difference, lifespan difference, and current similarity score. These features are selected because they directly reflect mechanical, economic, and semantic compatibility.

Feature	Type	Reason for Use
same_category	Binary	Ensures candidates are compared within the same major part category.
same_machine_family	Binary	Represents machine-level compatibility.
same_compatibility_group	Binary	Captures known or inferred substitute relationship.
same_function_type	Binary	Prevents functionally invalid recommendations.
same_material	Binary	Supports durability and usage-condition comparison.
diameter_diff	Numeric	Measures dimensional difference where applicable.
price_diff	Numeric	Supports cost-aware substitution.
lifespan_diff	Numeric	Supports durability-aware recommendation.
similarity_score	Numeric	Adds semantic similarity evidence.

Table 2.2: Pairwise features used for Random Forest compatibility prediction

2.7.3 Model Training and Integration

The dataset is split into training and testing subsets. The Random Forest classifier is trained on the training subset and evaluated on the testing subset using accuracy, precision, recall, and F1-score. After training, the model is saved and integrated with the backend recommender service. During recommendation generation, each candidate pair is passed to the prediction function, and the resulting probability is used as the ML component of the hybrid score.

2.8 Hybrid Recommendation Score

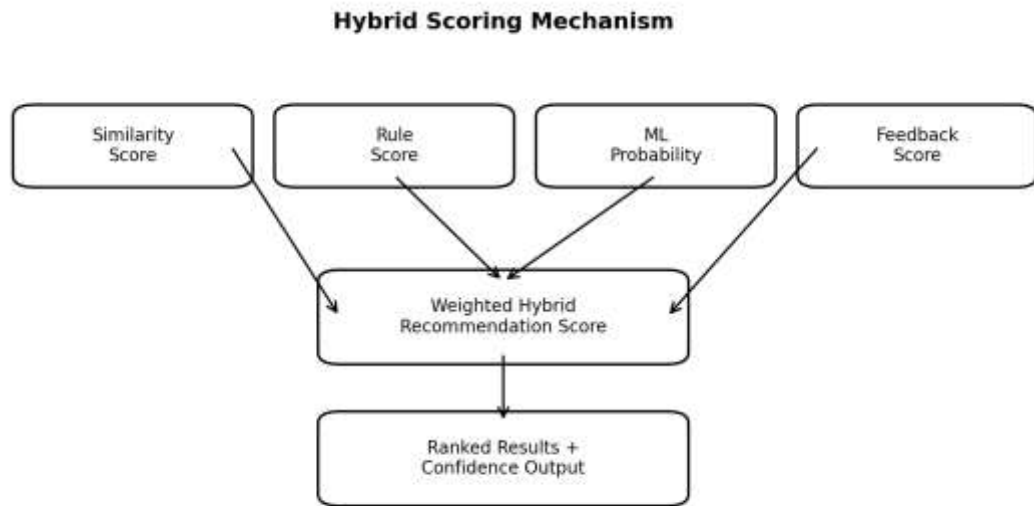


Figure 2.4: Hybrid scoring mechanism

The final recommendation score is calculated using a weighted combination of similarity score, rule score, Random Forest compatibility probability, and feedback score. The proposed formula is shown below:

$$\text{Final Score} = (0.45 \times \text{Similarity Score}) + (0.20 \times \text{Rule Score}) + (0.20 \times \text{RF Probability}) + (0.15 \times \text{Feedback Score})$$

Similarity receives the largest weight because it captures closeness between part descriptions and specifications after filtering. Rule score ensures that structured

compatibility conditions influence ranking. Random Forest probability adds learned compatibility evidence. Feedback score captures practical acceptance from previous user interactions. The weights can be tuned after experimental evaluation, but the proposed distribution provides a balanced starting point because no single factor is allowed to dominate the recommendation decision.

2.9 Explanation Builder

The explanation builder is designed to convert the recommendation decision into user-understandable text and structured fields. Each recommendation includes why_recommended, matched_fields, differences, substitute_level, confidence, and evidence_source. For example, a recommendation may state that the candidate is recommended because it belongs to the same machine family, same compatibility group, and same function type, with only a 1 mm diameter difference and a lower price. This explanation supports transparency and reduces user uncertainty.

Output Field	Example Value	Explanation
why_recommended	Same machine family and function type with close diameter match	Short natural-language reason.
matched_fields	machine_family, compatibility_group, function_type	Fields that matched between original and candidate.
differences	Diameter difference = 1 mm; price lower by 12%	Key differences that users should consider.
substitute_level	near	Replacement quality label.
confidence	0.84	Confidence score displayed to user.

Table 2.3: Explanation builder output structure

2.10 High-Level System Architecture

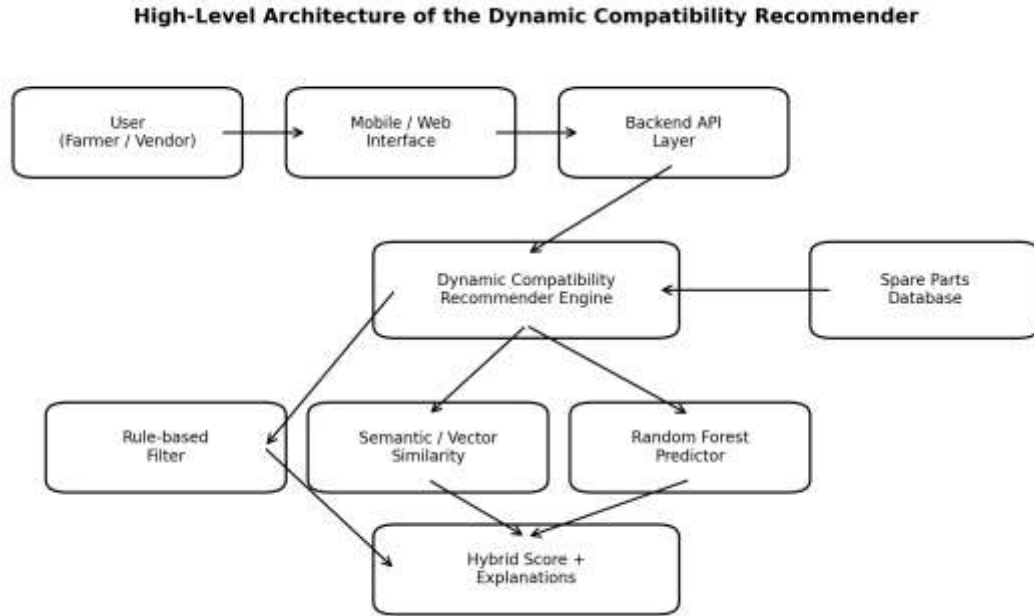


Figure 2.5: High-level architecture of the proposed component

The architecture follows a layered design. The user interacts with the frontend interface to select a spare part or search for a replacement. The request is sent to the backend API layer, which retrieves relevant part data and passes it to the Dynamic Compatibility Recommender Engine. The recommender engine applies hard rules, similarity ranking, Random Forest prediction, and hybrid scoring. The database stores part records, compatibility schema fields, feedback entries, and vector representations. The final recommendation response is returned to the frontend with compatible candidates, comparison details, confidence scores, and explanations.

2.11 Project Requirements Achieved

2.11.1 Functional Requirements

No.	Functional Requirement	Status	Description
1	Part selection and search	Achieved / To verify	User can select or search a spare part as the starting point.

2	Generate compatible recommendations	Achieved / To verify	System returns top recommended compatible spare parts.
3	Rule-based filtering	Achieved / To verify	Hard rules remove unsafe candidates.
4	Similarity ranking	Achieved / To verify	Vector similarity ranks relevant candidates.
5	Random Forest compatibility prediction	Achieved / To verify	ML probability is generated for candidate pairs.
6	Hybrid scoring	Achieved / To verify	Similarity, rules, ML, and feedback are combined.
7	Comparison view	Achieved / To verify	User can compare original and recommended part attributes.
8	Feedback capture	Achieved / To verify	User feedback is collected and used for reranking.
9	Explanation generation	Achieved / To verify	Recommendations include reasons and confidence values.

Table 2.4: Functional requirements

2.11.2 Non-Functional Requirements

Requirement	Description	Justification
Accuracy	Recommendations should be mechanically valid and relevant.	Achieved through rules, ML, and hybrid scoring.
Usability	Output should be understandable to farmers and vendors.	Explanation builder and comparison view support this.
Performance	Recommendations should be generated within acceptable response time.	Candidate filtering reduces unnecessary scoring.
Scalability	The system should support additional part categories.	Schema and modular architecture allow expansion.
Maintainability	The system should be easy to update.	Rules, model, API, and UI are separated.
Reliability	The system should avoid unsafe recommendations.	Hard rules protect against invalid candidates.
Security	User and vendor data should be	Authentication and API-level

	protected.	validation can be applied.
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Table 2.5: Non-functional requirements

2.12 Social, Security, Ethical Aspects and Limitations

2.12.1 Social Aspects

The component provides social value by helping farmers reduce the time and cost involved in finding suitable spare parts. In rural and semi-urban agricultural areas, access to expert technical advice may be limited. A compatibility recommender can support decision-making by providing structured alternatives and explanations. Vendors also benefit because they can recommend substitutes more confidently and improve customer satisfaction.

2.12.2 Security Aspects

The system should protect user information, vendor records, and feedback data. API endpoints should validate input to prevent incorrect or malicious requests. If user accounts are used, authentication and authorization should be implemented. Sensitive data such as user contact details, vendor stock levels, and transaction records should not be exposed unnecessarily through API responses.

2.12.3 Ethical Aspects

Ethical responsibility is important because incorrect recommendations can cause financial loss or machinery damage. Therefore, the system should avoid presenting uncertain substitutes as guaranteed exact replacements. Confidence scores, substitute levels, and evidence sources must be shown clearly. The system should also avoid bias toward a particular vendor or brand unless supported by compatibility and user requirements.

2.12.4 Limitations

The main limitation is dependency on data quality and availability. If part specifications are incomplete or incorrect, recommendation accuracy may decrease. Another limitation is the size of the labelled pairwise dataset used for Random Forest training. A larger

dataset with real field validation would improve model reliability. The system also requires expert validation for critical compatibility rules, especially for high-risk mechanical components.

2.13 Commercialization Aspects of the Product

The Dynamic Compatibility Recommender has strong commercialization potential because it addresses a practical problem in agricultural machinery maintenance. The target users include farmers, spare-part vendors, machinery service centers, repair workshops, and agricultural equipment suppliers. The component can be commercialized as part of a mobile application, a vendor dashboard, or an API integrated into an agricultural marketplace.

A subscription model can be offered to spare-part vendors who need advanced recommendation and inventory insights. Farmers can access basic recommendations for free while vendors pay for premium features such as stock-aware recommendation, substitute analytics, and compatibility data management. Another possible model is a marketplace commission model, where compatible recommendations lead to vendor sales. The system can also be licensed to machinery service centers and equipment distributors.

The value proposition is based on reducing downtime, reducing incorrect purchases, improving customer satisfaction, and increasing spare-part discoverability. In the long term, the compatibility database can become a valuable digital asset for agricultural machinery maintenance in Sri Lanka and similar agricultural economies.

2.14 Testing and Implementation

2.14.1 Implementation Summary

The implementation consists of backend APIs, database schema updates, recommendation logic, Random Forest model integration, and frontend display. The backend exposes endpoints for part search, recommendation generation, comparison view, feedback submission, vector rebuild, and model prediction. The database stores

part details and compatibility schema fields. The ML model is trained separately and loaded by the backend service for probability prediction. The frontend displays recommendations, comparison details, substitute labels, and confidence scores.

2.14.2 Test Case Summary

Test Case	Input	Expected Output	Status
TC01 - Parts API test	Request all parts	Valid part list returned	To be completed with final evidence
TC02 - Recommendation API test	Selected part ID	Top compatible recommendations returned	To be completed with final evidence
TC03 - Hard filter test	Candidate with different function type	Candidate excluded	To be completed with final evidence
TC04 - Similarity ranking test	Valid candidate set	Candidates ranked by similarity	To be completed with final evidence
TC05 - RF prediction test	Part A and Part B pair	Compatibility probability returned	To be completed with final evidence
TC06 - Hybrid score test	Candidate with all scores	Final score calculated correctly	To be completed with final evidence
TC07 - Comparison flow	Open recommended part comparison	Differences displayed	To be completed with final evidence
TC08 - Feedback flow	Submit feedback	Feedback saved and reranking updated	To be completed with final evidence

Table 2.6: Test case summary

2.15 Tools and Technologies

Technology / Tool	Purpose	Justification
Backend framework	API development and recommendation services	Separates business logic from frontend and supports integration.
Database system	Storage of part data, schema fields, feedback, and vectors	Required for structured retrieval and compatibility management.
Python	Machine learning development	Provides strong ML ecosystem for data preprocessing and model training.
Scikit-learn	Random Forest classifier and evaluation metrics	Well-established library for supervised machine learning.
Frontend framework	User interface for search, recommendations, comparison,	Provides interactive user experience.

	and feedback	
JSON / REST APIs	Data exchange between frontend and backend	Supports modular communication.
Version control	Source-code management	Improves team collaboration and maintainability.

Table 2.7: Tools and technologies

3. RESULTS AND DISCUSSION

3.1 Evaluation Design

The evaluation of the Dynamic Compatibility Recommender is divided into recommender ranking evaluation and machine learning classifier evaluation. Ranking evaluation measures whether relevant compatible parts appear in the top recommendation positions. Classifier evaluation measures whether the Random Forest model correctly predicts pairwise compatibility. The planned recommender variants are rules-only, similarity-only, similarity with feedback, Random Forest-assisted, and final hybrid model. This comparison is required to demonstrate that the proposed hybrid method improves over simpler baselines.

3.2 Recommendation Performance Results

Table 3.1 is prepared for final measured recommendation results. The values should be filled using actual test data after running the final evaluation script. No unmeasured values should be inserted into the final submission because fabricated results would not be academically valid.

Recommender Variant	Precision@5	Recall@5	Observation
Rules only			Safe but may miss semantically related substitutes.
Similarity only			Flexible but may include invalid candidates without rules.
Similarity + feedback			Improves ranking using practical acceptance data.
RF-assisted			Adds learned compatibility probability.
Final hybrid			Expected to perform best because it combines all evidence sources.

Table 3.1: Recommendation performance comparison template

3.3 Random Forest Classifier Results

Table 3.2 is prepared for the final Random Forest classifier results. Accuracy measures overall correctness, precision measures how many predicted compatible pairs are truly compatible, recall measures how many actual compatible pairs are detected, and F1-score provides a balance between precision and recall. These values must be generated from the final test split of the pairwise dataset.

Metric	Measured Value	Interpretation
Accuracy		Overall pairwise classification correctness.
Precision		Reliability of predicted compatible pairs.
Recall		Ability to identify actual compatible pairs.
F1-score		Balanced classification performance.

Table 3.2: Random Forest classifier performance template

3.4 Research Findings

The expected research finding is that a hybrid recommendation approach is more appropriate for agricultural spare-part compatibility than a single-method recommender. Rule-based filtering is necessary to protect mechanical validity. Similarity ranking is useful for flexible discovery of alternatives. Random Forest prediction adds learned evidence from labelled compatibility pairs. Feedback-weighted reranking improves practical relevance over time. Explanation generation improves transparency and user trust. Together, these findings support the main claim that compatibility recommendation in agricultural machinery should be treated as a structured, explainable, and hybrid decision problem.

3.5 Discussion

The proposed component improves spare-part recommendation in three main ways. First, it introduces structured compatibility fields that make mechanical relationships

explicit. Second, it combines multiple decision factors rather than depending on only similarity or only rules. Third, it provides explanation output so that users can understand why a recommended candidate is suitable. This is particularly important in agricultural settings where users may rely on the system to avoid costly mistakes.

The design also supports future scalability. New part categories can be added by defining compatibility groups and function types. Additional machine learning models can be tested later if more data becomes available. Graph-based reasoning using technologies such as Neo4j can also be explored in future work to represent complex relationships among machines, parts, substitutes, vendors, and feedback records.

3.6 Summary of Individual Contribution

Area	Individual Contribution
Compatibility schema	Designed machine family, compatibility group, function type, substitute level, confidence score, and evidence source fields.
Rule system	Defined hard rules and soft rules for compatibility-safe recommendation.
Similarity alignment	Connected semantic similarity with new compatibility schema fields.
Random Forest model	Planned and integrated pairwise compatibility prediction workflow.
Hybrid score	Designed weighted scoring using similarity, rules, RF probability, and feedback.
Explanation builder	Designed output fields for why_recommended, matched_fields, differences, substitute_level, and confidence.
Evaluation plan	Prepared recommender and classifier evaluation using Precision@5, Recall@5, Accuracy, Precision, Recall, and F1-score.
UI integration	Connected recommendations, comparison, feedback, and inventory-aware output requirements to the frontend flow.

Table 3.3: Individual contribution summary

4. CONCLUSION

This research focused on the design and development of a Dynamic Compatibility Recommender for the Smart Spare Parts Management System for Agricultural Machinery. The component was designed to solve the practical problem of identifying compatible and functionally equivalent spare parts when original parts are unavailable, expensive, or difficult to identify. The proposed approach improves upon simple similarity-based recommendation by introducing a structured compatibility schema, hard and soft rule filtering, semantic similarity ranking, Random Forest compatibility prediction, feedback-aware hybrid scoring, and explanation generation.

The structured compatibility schema allows the system to represent machine family, compatibility group, function type, substitute level, confidence score, and evidence source. These fields are essential for transforming a basic recommendation system into a compatibility-aware recommender. The rule-based layer protects the system from invalid recommendations, while the similarity layer supports flexible discovery of related candidates. The Random Forest classifier provides predictive compatibility evidence, and the feedback score allows practical user acceptance to influence future ranking. The explanation builder improves transparency by showing matched fields, differences, confidence, and substitute level.

The component contributes both practical and research value. Practically, it can help farmers reduce downtime and avoid incorrect spare-part purchases. Vendors can use it to recommend substitutes more confidently and manage customer requests effectively. From a research perspective, it demonstrates how hybrid recommendation can be adapted to a technical agricultural domain where compatibility, explainability, and data quality are critical. Future work can include expanding the dataset, validating compatibility relationships with domain experts, improving the ML model with larger labelled data, applying graph-based reasoning, and integrating real-time inventory and vendor availability.

5. REFERENCES

- [1] G. Adomavicius and A. Tuzhilin, “Toward the next generation of recommender systems: A survey of the state-of-the-art and possible extensions,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 17, no. 6, pp. 734-749, 2005.
- [2] F. Ricci, L. Rokach, and B. Shapira, Eds., *Recommender Systems Handbook*, 2nd ed. New York, NY, USA: Springer, 2015.
- [3] R. Burke, “Hybrid recommender systems: Survey and experiments,” *User Modeling and User-Adapted Interaction*, vol. 12, no. 4, pp. 331-370, 2002.
- [4] R. Burke, “Knowledge-based recommender systems,” in *Encyclopedia of Library and Information Systems*, New York, NY, USA: Marcel Dekker, 2000.
- [5] G. Salton, A. Wong, and C. S. Yang, “A vector space model for automatic indexing,” *Communications of the ACM*, vol. 18, no. 11, pp. 613-620, 1975.
- [6] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge, U.K.: Cambridge University Press, 2008.
- [7] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5-32, 2001.
- [8] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. New York, NY, USA: Springer, 2009.
- [9] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830, 2011.
- [10] Y. Zhang and X. Chen, “Explainable recommendation: A survey and new perspectives,” *Foundations and Trends in Information Retrieval*, vol. 14, no. 1, pp. 1-101, 2020.
- [11] J. L. Herlocker, J. A. Konstan, L. G. Terveen, and J. T. Riedl, “Evaluating collaborative filtering recommender systems,” *ACM Transactions on Information Systems*, vol. 22, no. 1, pp. 5-53, 2004.
- [12] A. Gunawardana and G. Shani, “Evaluating recommender systems,” in *Recommender Systems Handbook*, F. Ricci, L. Rokach, and B. Shapira, Eds. New York, NY, USA: Springer, 2015, pp. 265-308.
- [13] D. M. W. Powers, “Evaluation: From precision, recall and F-measure to ROC, informedness, markedness and correlation,” *Journal of Machine Learning Technologies*, vol. 2, no. 1, pp. 37-63, 2011.
- [14] P. Lops, M. de Gemmis, and G. Semeraro, “Content-based recommender systems: State of the art and trends,” in *Recommender Systems Handbook*, F. Ricci, L. Rokach, B. Shapira, and P. B. Kantor, Eds. Boston, MA, USA: Springer, 2011, pp. 73-105.
- [15] C. C. Aggarwal, *Recommender Systems: The Textbook*. Cham, Switzerland: Springer, 2016.

- [16] Y. Koren, R. Bell, and C. Volinsky, "Matrix factorization techniques for recommender systems," *Computer*, vol. 42, no. 8, pp. 30-37, 2009.
- [17] P. Resnick, N. Iacovou, M. Suchak, P. Bergstrom, and J. Riedl, "GroupLens: An open architecture for collaborative filtering of netnews," in *Proc. ACM Conf. Computer Supported Cooperative Work*, 1994, pp. 175-186.
- [18] B. Sarwar, G. Karypis, J. Konstan, and J. Riedl, "Item-based collaborative filtering recommendation algorithms," in *Proc. 10th International World Wide Web Conference*, 2001, pp. 285-295.
- [19] A. I. Schein, A. Popescul, L. H. Ungar, and D. M. Pennock, "Methods and metrics for cold-start recommendations," in *Proc. 25th Annual International ACM SIGIR Conference*, 2002, pp. 253-260.
- [20] M. J. Pazzani and D. Billsus, "Content-based recommendation systems," in *The Adaptive Web*, P. Brusilovsky, A. Kobsa, and W. Nejdl, Eds. Berlin, Germany: Springer, 2007, pp. 325-341.
- [21] D. Jannach, M. Zanker, A. Felfernig, and G. Friedrich, *Recommender Systems: An Introduction*. Cambridge, U.K.: Cambridge University Press, 2010.
- [22] Food and Agriculture Organization of the United Nations, *Mechanization for Rural Development: A Review of Patterns and Progress from Around the World*. Rome, Italy: FAO, 2013.
- [23] Food and Agriculture Organization of the United Nations, *Sustainable Agricultural Mechanization: A Framework for Africa*. Addis Ababa, Ethiopia: FAO, 2018.
- [24] World Bank Group, *Enabling the Business of Agriculture 2019*. Washington, DC, USA: World Bank, 2019.
- [25] E. A. Silver, D. F. Pyke, and D. J. Thomas, *Inventory and Production Management in Supply Chains*, 4th ed. Boca Raton, FL, USA: CRC Press, 2016.
- [26] S. Chopra and P. Meindl, *Supply Chain Management: Strategy, Planning, and Operation*, 6th ed. Boston, MA, USA: Pearson, 2016.
- [27] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021.
- [28] R. G. Brown, *Statistical Forecasting for Inventory Control*. New York, NY, USA: McGraw-Hill, 1959.
- [29] J. D. Croston, "Forecasting and stock control for intermittent demands," *Operational Research Quarterly*, vol. 23, no. 3, pp. 289-303, 1972.
- [30] A. A. Syntetos and J. E. Boylan, "The accuracy of intermittent demand estimates," *International Journal of Forecasting*, vol. 21, no. 2, pp. 303-314, 2005.
- [31] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Waltham, MA, USA: Morgan Kaufmann, 2012.

- [32] M. Kuhn and K. Johnson, *Applied Predictive Modeling*. New York, NY, USA: Springer, 2013.
- [33] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.
- [34] M. de Gemmis, P. Lops, G. Semeraro, and P. Basile, “Integrating tags in a semantic content-based recommender,” in *Proc. ACM Conference on Recommender Systems*, 2008, pp. 163-170.
- [35] J. Bobadilla, F. Ortega, A. Hernando, and A. Gutiérrez, “Recommender systems survey,” *Knowledge-Based Systems*, vol. 46, pp. 109-132, 2013.
- [36] A. Felfernig and R. Burke, “Constraint-based recommender systems: Technologies and research issues,” in *Proc. 10th International Conference on Electronic Commerce*, 2008, pp. 1-10.
- [37] A. Felfernig, G. Friedrich, D. Jannach, and M. Zanker, “An integrated environment for the development of knowledge-based recommender applications,” *International Journal of Electronic Commerce*, vol. 11, no. 2, pp. 11-34, 2006.
- [38] J. McAuley, C. Targett, Q. Shi, and A. van den Hengel, “Image-based recommendations on styles and substitutes,” in *Proc. 38th International ACM SIGIR Conference*, 2015, pp. 43-52.
- [39] R. T. Fielding, “Architectural styles and the design of network-based software architectures,” Ph.D. dissertation, University of California, Irvine, CA, USA, 2000.
- [40] ISO/IEC 25010:2011, *Systems and Software Engineering - Systems and Software Quality Requirements and Evaluation (SQuaRE) - System and Software Quality Models*, International Organization for Standardization, 2011.

APPENDIX A: SAMPLE RECOMMENDATION API RESPONSE

The following sample response shows the type of structured output expected from the Dynamic Compatibility Recommender. The values should be adapted to the final backend implementation.

```
{
  "selected_part_id": "P001",
  "recommendations": [
    {
      "part_id": "P014",
      "name": "A-Type V-Belt A-54",
      "machine_family": "Small Agricultural Machinery",
      "compatibility_group": "VBELT_A_SECTION_54",
      "function_type": "Power Transmission",
      "substitute_level": "near",
      "similarity_score": 0.87,
      "rule_score": 0.92,
      "rf_probability": 0.84,
      "feedback_score": 0.76,
      "final_score": 0.85,
      "confidence": 0.84,
      "why_recommended": "Same function type and compatibility group with close specification match.",
      "matched_fields": ["machine_family", "compatibility_group", "function_type"],
      "differences": ["price lower by 12%", "lifespan difference: 3 months"]
    }
  ]
}
```

APPENDIX B: SAMPLE COMPATIBILITY DATA RECORDS

Part	Machine Family	Compatibility Group	Function Type	Substitute Level	Confidence	Evidence Source
Rear Tractor Tyre 13.6-	Medium Tractor Series 45-60 HP	TYRE_REAR_13_6_28	Traction and Load Bearing	exact	0.95	Manufacturer specification

28						
Rear Tractor Tyre 14.9-28	Medium Tractor Series 50-65 HP	TYRE_REAR_13_6_28	Traction and Load Bearing	near	0.82	Expert recommendation
V-Belt A-54	Small Agricultural Machinery	VBELT_A_SECTION_54	Power Transmission	exact	0.93	Standard size matching
V-Belt B-52	Small Agricultural Machinery	VBELT_A_SECTION_54	Power Transmission	functional	0.68	Engineering judgement
Oil Filter OF-123	Diesel Tractor Engines 30-50 HP	OIL_FILTER_STD_SM ALL	Lubrication Filtration	exact	0.97	OEM cross-reference

Table B.1: Sample compatibility records